

# STEP UP

## User Manual

The original MSX game, wrapped in a modern layer



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# 1 Welcome to Step Up

*Step Up* is an action platform game originally released for MSX in 1983 and developed by Takara and Marvel Soft. You play a lone astronaut stranded on a hostile planet. His only chance of returning home is to climb to the rooftop and reach the waiting spaceship.

**MSX StepUp HybridHLE preserves the original cartridge program.** This is not a remake that imitates its rules: the original Z80 instructions still decide movement, collisions, enemies, scoring, lives, and progression between stages. Godot acts as a modern outer layer that runs and presents that program on Windows, Linux, macOS, and Android.

**Important.** The original cartridge is not distributed with this project. To play, you need a legally obtained dump of your own *Step Up* cartridge.

## 2 Getting started

### 2.1 First launch

1. Open **MSX StepUp HybridHLE**.
2. When the setup screen appears, select **Select Step Up ROM**.
3. Choose your cartridge's `.rom` file. If it is inside a ZIP archive, extract it first.
4. The application accepts 8, 16, or 32 KiB dumps, extracts the valid 8 KiB bank, and stores a private copy for future launches.

On desktop computers, you may instead create a `roms` directory beside the executable and save the file as `roms/stepup.rom`. On macOS, place the directory beside the `.app` bundle, not inside it.

### 2.2 Starting a game

An opening illustration is displayed for a few seconds after the game loads. Press **Space** —or **ACTION** on Android— to skip it. On the title screen, use the action button again to start when prompted by the game.

## 3 Game objective

The building has ten floors, marked F1 through F10 and connected by blue ladders. Guide the astronaut from the lower area to the rooftop while avoiding the creatures and moving hazards patrolling each floor.

Once you reach the roof, a small spaceship appears. **Reaching the top is not enough: you must jump aboard before it flies away.** Succeed and you advance to the next stage; wait too long and you will be left exposed to the enemies on the rooftop.

### 3.1 Hazards

Each threat behaves differently. You will encounter:

- mice or rats that scurry along the platforms;

- bats that fly vertically;
- monsters that move along ladders and floors;
- spiders that periodically emerge from their webs;
- moving obstacles such as the red girder.

Contact can knock the astronaut to the ground and costs one life. After losing a life, you start again from the bottom. A game begins with three lives, and an extra life is awarded every 10,000 points.

### 3.2 Energy barrier

The astronaut has temporary protection. While it is active, the character turns red and consumes part of the energy gauge at the bottom of the screen. The supply is restored at the beginning of every new life or stage; if it runs out, the character is defeated.

Later stages retain the same objective but increase the pressure and the number of hazards; for example, more spider webs appear.

## 4 Controls

| Action                               | Desktop | Android      |
|--------------------------------------|---------|--------------|
| <b>Walk left or right</b>            | ← / →   | LEFT / RIGHT |
| <b>Climb up or down</b>              | ↑ / ↓   | UP / DOWN    |
| <b>Jump or perform an action</b>     | Space   | ACTION       |
| <b>Skip the opening illustration</b> | Space   | ACTION       |

On Android, five buttons are overlaid on the display. The four directions are on the left, with **ACTION** on the right.

### 4.1 Playing tips

- Watch the route and rhythm of each enemy before moving.
- Wait for a safe opening before entering a ladder: your options for evading an enemy are limited while climbing.
- Save the energy barrier for difficult crossings.
- Once on the roof, get into position and jump aboard without delay.
- Do not rush. *Step Up* rewards memory and timing more than constant speed.

## 5 What Hybrid HLE means

In this project, **Hybrid HLE** means combining low-level CPU emulation with high-level replacements for the specific MSX services required by the game.

Keyboard or touchscreen → MSX keyboard matrix → original Z80 code  
 → memory and VRAM → image and sound presented by Godot

The project's native extension emulates the Z80 microprocessor and executes the cartridge instructions. At startup, it assembles the 64 KiB address space visible to that CPU:

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|           |                                              |
|-----------|----------------------------------------------|
| 0000-3FFF | C-BIOS for MSX1                              |
| 4000-5FFF | player-provided <i>Step Up</i> cartridge     |
| 6000-7FFF | cartridge mirror                             |
| 8000-FFFF | captured RAM state at the game's entry point |

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Godot surrounds this virtual machine with modern components:

- it converts keyboard and touch controls into the keyboard matrix expected by an MSX;
- it interprets operations sent to the TMS9918A video hardware and renders its tiles and sprites through the engine;
- it observes events generated by the original program and plays renewed sound effects;
- it provides windowing, scaling, file selection, and integration with current operating systems.

This is why the result is genuinely *Step Up*: the original Z80 code retains authority over the game, while the HLE layer replaces only the hardware and services around it. It is not intended to be a general-purpose MSX emulator capable of loading any cartridge; it is tailored specifically to this title.

## 6 Troubleshooting

### The application asks for the ROM again

Make sure that you selected a recognizable 8, 16, or 32 KiB dump. The file selector cannot open ZIP archives; extract the `.rom` file first.

### The executable will not open on Windows or macOS

Community builds may be unsigned. On Windows, extract the entire ZIP before running the executable and review the SmartScreen warning. On macOS, you may need to Control-click the application and choose **Open**.

### Linux reports that the file is not executable

Grant permission once from a terminal:

```
chmod +x MSX-StepUp-HybridHLE.x86_64
```

## The controls do not respond

Click inside the window to give it focus. Use the cursor keys rather than WASD, and press **Space** to jump or confirm.

## 7 Credits, rights, and sources

MSX StepUp HybridHLE is an independent preservation and porting project created in 2026 by **DiHalt**, a team formed by Fernando García and Clemente Tiñena. On this game, Fernando is responsible for the Hybrid HLE implementation, integration, and project artwork; Clemente is responsible for sound and replacement audio. The project is not affiliated with or endorsed by the original developers or publishers.

The project's original code is distributed under GPL-3.0-or-later. Original project assets are licensed under CC BY-SA 4.0, subject to the qualifications in `ASSET-LICENSE.md`. *Step Up*, its cartridge, its materials, and elements derived from its packaging belong to their respective rights holders.

The historical and gameplay descriptions were checked on 12 August 2026 against:

- [MobyGames: Step Up](#), game overview and description;
- [Generation MSX: Step Up](#), technical and historical entry;
- [MSX Computing, August/September 1985, p. 54](#), a contemporary review describing floors, enemies, the energy barrier, lives, and stage progression.

The controls and technical explanation describe the current implementation of MSX StepUp HybridHLE directly.

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Climb, time your move, and do not let the spaceship get away.